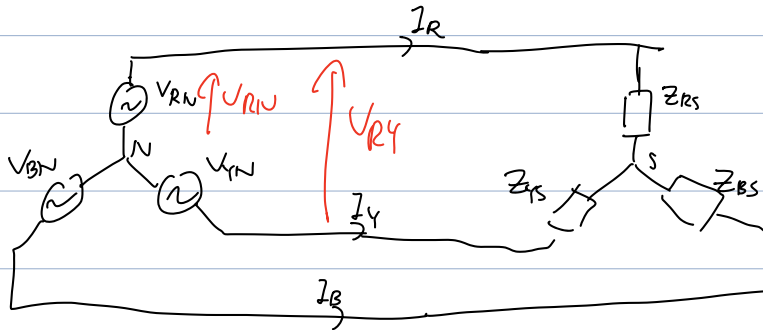


Q.1.

(a) Circuit diagram :



(i) $Z_{RS} = Z_{YS} = Z_{BS} = Z = 139 + j56 = 149.86 \angle 21.9^\circ \Omega$

$$I_R = \frac{V_{RN}}{Z_{RS}} = \frac{11 \angle 0^\circ / \sqrt{3}}{149.86 \angle 21.9^\circ} = 73.4 \angle -21.9^\circ \text{ A} / \sqrt{3}$$

$$I_Y = \frac{V_{YN}}{Z_{YS}} = \frac{11 \angle -120^\circ / \sqrt{3}}{149.86 \angle 21.9^\circ} = 73.4 \angle -141.9^\circ \text{ A} / \sqrt{3}$$

$$I_B = \frac{V_{BN}}{Z_{BS}} = \frac{11 \angle 120^\circ / \sqrt{3}}{149.86 \angle 21.9^\circ} = 73.4 \angle 98.1^\circ \text{ A} / \sqrt{3}$$

(ii) Total real power : $P_{\text{total}} = 3 |I_R| \cdot |V_{ph}| \cdot \cos \theta$

$$= 3 \times 73.4 \times 11 \angle 0^\circ \cdot \cos 21.9^\circ$$

$$= 2.25 \text{ MW} / 3 = 750 \text{ kW}$$

0.81 kVA

Total apparent power : $S_{\text{total}} = 3 |I_R| \cdot |V_{ph}| = 2.42 \text{ MVA}$

Total reactive power : $Q_{\text{total}} = 3 |I_R| \cdot |V_{ph}| \cdot \sin \theta = 0.40 \text{ MVar}$

0.30 MVar

Power factor $PF = \cos \theta = 0.928$

(b). $Z_{RS} = 100 + j40 = 107.7 \angle 21.8^\circ \Omega$

$$Z_{YS} = 250 + j80 = 262.5 \angle 17.7^\circ \Omega$$

$$Z_{BS} = 149.86 \angle 21.9^\circ \Omega$$

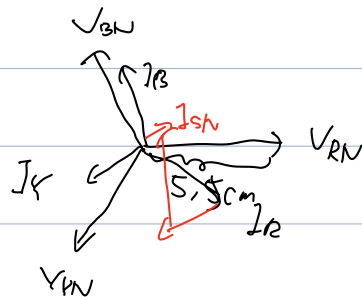
Therefore:

$$I_R = \frac{11k \angle 0^\circ / \sqrt{3}}{107.7 \angle 21.8^\circ} = 102.1 \angle -21.8^\circ \text{ A} / \sqrt{3}$$

$$I_Y = \frac{11k \angle -120^\circ / \sqrt{3}}{262.5 \angle 17.7^\circ} = 41.9 \angle -137.7^\circ \text{ A} / \sqrt{3}$$

$$I_B = \frac{11k \angle 120^\circ / \sqrt{3}}{149.86 \angle 21.9^\circ} = 73.4 \angle 98.1^\circ \text{ A} / \sqrt{3}$$

The phasor diagram



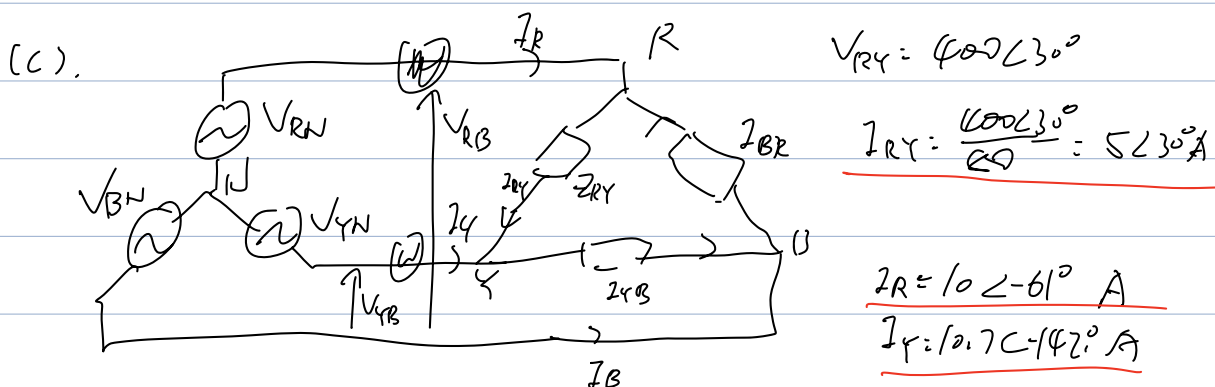
$$20A = 1cm$$

$$10KV = 5cm$$

$$(iii) \quad I_{SN} = I_{RS} + I_{YS} + I_{BS} = 53.9 \angle 7.0^\circ \text{ A}$$

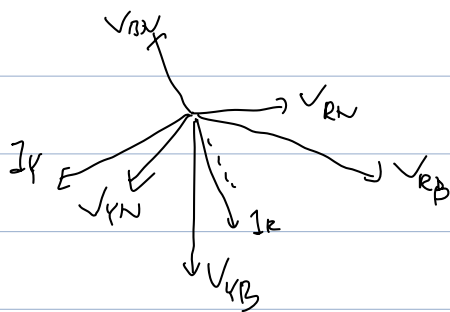
$2.7cm \quad 7.0^\circ$

$$\pi \left(\frac{d}{2}\right)^2 = \frac{|I|}{J} = \frac{53.9 \text{ A}}{5A/mm^2} \Rightarrow d = 2\sqrt{\frac{|I|}{J\pi}} = 3.7 \text{ mm}$$



$$I_R + I_Y + I_B = 0 \Rightarrow I_B = -I_R - I_Y = -10 \angle (-30^\circ - 31^\circ) - 10.7 \angle (-90^\circ - 51^\circ)$$

$$= 3.58 + j15.33$$



$$= 15.75 \angle 76.8^\circ \text{ A}$$

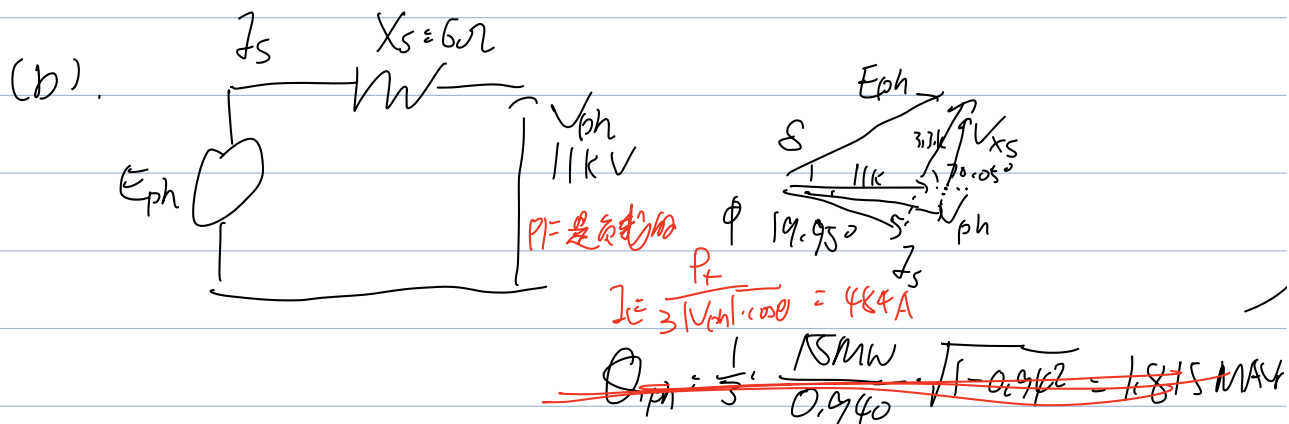
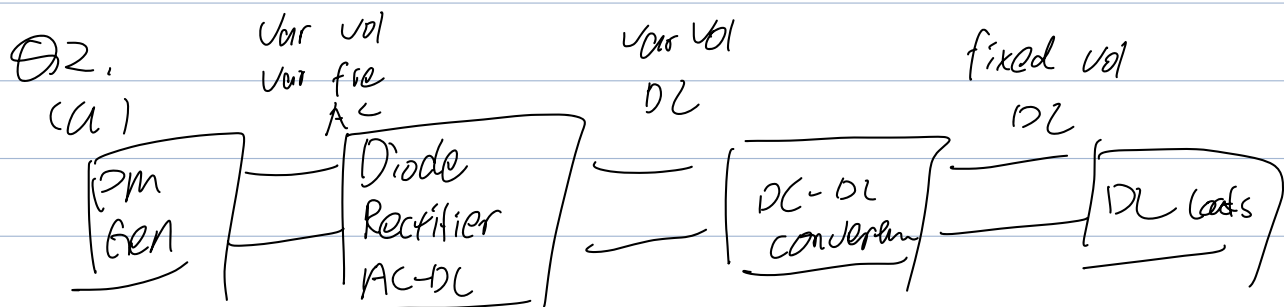
$$\begin{aligned} I_{RY} - I_{BR} &= I_R \Rightarrow I_{BR} = I_{RY} - I_R = 11.26 \angle 92.6^\circ \text{ A} \\ I_{YB} - I_{RY} &= I_Y \Rightarrow I_{YB} = I_{RY} + I_Y = 5.79 \angle 15^\circ \text{ A} \\ I_{BR} - I_{YB} &= I_B \end{aligned}$$

$$RY: P_{RY} = \frac{400^2}{80} = 2 \text{ kW}$$

$$YB: P_{YB} = |V_{YB}| \cdot |I_{YB}| \cdot \cos \theta = 400 \cdot 5.79 \cdot \cos(45^\circ) = 1.64 \text{ kW}$$

$$BR: P_{BR} = |V_{BR}| \cdot |I_{BR}| \cdot \cos \theta = 400 \cdot 11.26 \cdot \cos(150 - 92.6) = 2.48 \text{ kW}$$

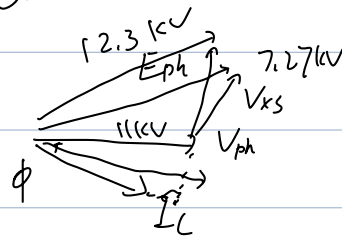
$$P_{\text{total}} = 3.45 \text{ kW} + 2.64 \text{ kW} = 6.07 \text{ kW}$$



$$V_{XS} = jI_L \cdot X_S = 2.9 \text{ kV} \angle 70^\circ \quad V_{XS} = \sqrt{S_{ph}} \cdot X_S = 3.3 \text{ kV} \quad 2.9 \text{ kV}$$

$$E_{ph} = V_{ph} + V_{XS} = 12.52 \angle 4.3^\circ \text{ kV} \\ 12.3 \angle 13^\circ \text{ kV}$$

(c) grid

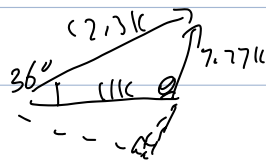


V_{ph} fixed

$$V_{ph} = 11 \text{ kV}$$

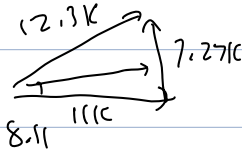
$$I_L = \frac{P_e/3}{V_{ph}} = 1.212 \times 10^3 \text{ A}$$

$$|V_{XS}| = I_L \cdot X_S = 7.27 \times 10^3 \text{ V}$$



$$7.27^2 = 12.3^2 + 11^2 - 2 \times 12.3 \times 11 \times \cos \delta$$

$$\cos \delta = 0.8109 \Rightarrow \delta = 35.4^\circ \approx 36^\circ$$

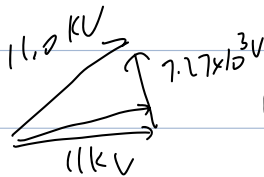


$$12.3^2 = 11^2 + 7.27^2 - 2 \times 11 \times 7.27 \times \cos \theta$$

$$\theta = 81.8^\circ$$

$$\Rightarrow -\cos \theta = \text{pf} = (-0.11) = 0.99 \text{ leading}$$

(d)



$$E_{ph} = 11.0 \text{ kV}$$

Q5. (a)

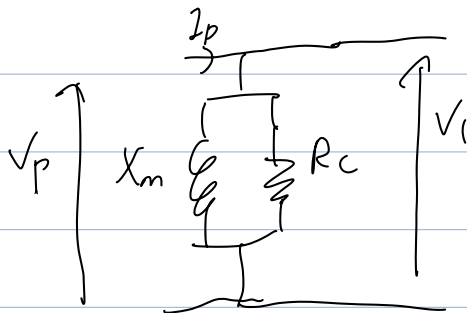
(i) $V_{out} \downarrow$ losses $\uparrow$ regulation $\uparrow$

$I \uparrow$

$$V_{max} = \frac{2\pi}{T} N \omega B_{max} A_c$$

(ii) . . .

(b) $TR = 230:24 = 9.58$

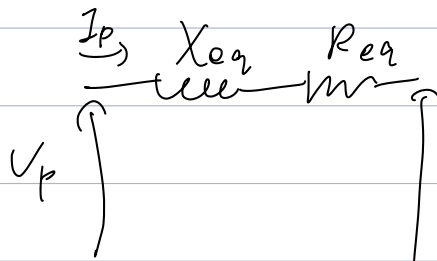


$$P_p = \frac{V_p^2}{R_c} \Rightarrow R_c = \frac{V_p^2}{P_p} = 1058 \Omega$$

$$S_p = I_1 V_p = 200.1 \text{ VA}$$

$$Q_p = \sqrt{S_p^2 - P_p^2} = 193.8 \text{ VAR}$$

$$Q_p = \frac{V_p^2}{X_m} \Rightarrow X_m = \frac{V_p^2}{Q_p} = 273 \Omega$$



$$P_p = I_1^2 R_{eq} \Rightarrow R_{eq} = \frac{P_p}{I_1^2} = 1.6 \Omega$$

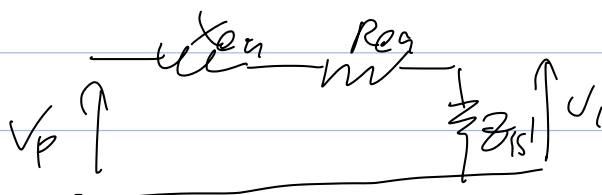
$$S_p = V_p I_1 = 85 \text{ VA}$$

$$Q_p = \sqrt{S_p^2 - P_p^2} = 75 \text{ VAR}$$

$$Q_p = I_1^2 X_{eq} \Rightarrow X_{eq} = \frac{Q_p}{I_1^2} = 3 \Omega$$

(c) (let copper loss = iron loss, i.e.)

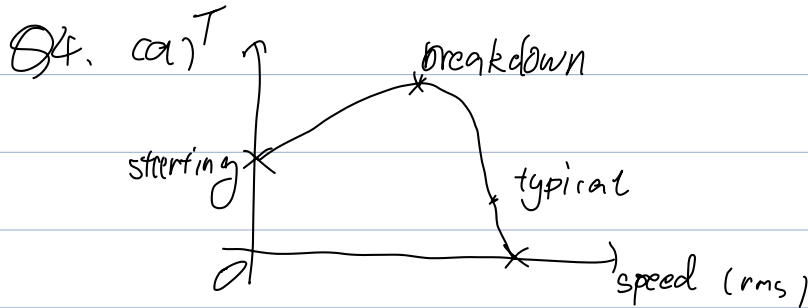
$$\frac{V_p^2}{R_c} = I_1^2 R_{eq} \Rightarrow I_1 = \frac{V_p}{\sqrt{R_c R_{eq}}} = 5.54 \text{ A}$$



$$I_1 = \frac{V_p}{\sqrt{X_{eq}^2 + R_{eq}^2 + Z_{is}^2}}$$

$$\Rightarrow Z_{is} = \sqrt{\frac{V_p^2}{I_1^2} - X_{eq}^2 - R_{eq}^2} = 3.54 \Omega$$

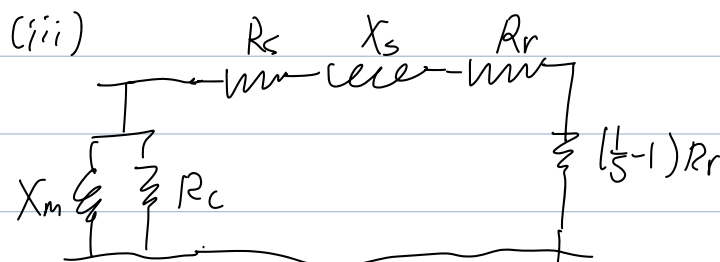
$$Z_s = \frac{Z_{k'1}}{I_{k2}} = 0.432$$



$$\begin{aligned} \text{(b)(i)} P_{in} &= 3 \times |V_{ph}| \cdot |I_L| \cdot \cos \theta \\ &= 3 \times 110 \times 16.2 \times 0.938 \\ &= 5.00 \text{ kW} \end{aligned}$$

$$\begin{aligned} \text{(ii)} N_{sy} &= \frac{60 f_s}{P/2} = \frac{60 \times 60}{3} = 1200 \text{ rpm} \\ \text{From } s &= \frac{N_{sy} - N_r}{N_{sy}} \times 100\% \end{aligned}$$

$$\Rightarrow N_r = N_{sy}(1-s) = 1200 \times 0.97 = 1164 \text{ rpm}$$



$$\text{no load loss} \Rightarrow \frac{V_{ph}^2}{R_c} = 220 \text{ W}$$

$$\text{total stator loss} = 3 I_s^2 \cdot R_s = 180 \text{ W}$$

$$\Rightarrow P_{gap} = P_{in} - P_{no-load} - P_s = 4.60 \text{ kW}$$

$$P_{mech} = P_{gap}(1-s) = 4.46 \text{ kW}$$

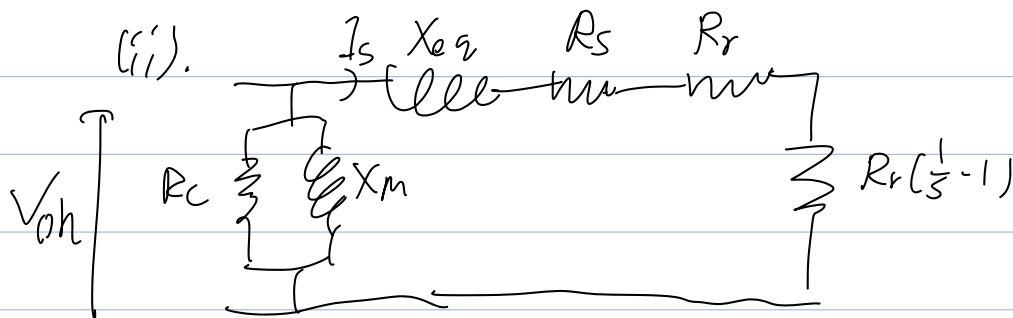
$$T = \frac{P_{mech}}{\omega_r} = \frac{P_{mech}}{N_r \cdot \frac{2\pi}{60}} = 36.59 \text{ Nm}$$

$$(iv) P_{mech} = P_{mech} \cdot s = 0.138 \text{ kW} = 138 \text{ W}$$

$$(v) \text{ Efficiency} = \frac{P_{mech}}{P_{in}} \times 100\% = \frac{4.46}{5.00} = 89.2\%$$

$$cc), (i) N_{sy} = \frac{60f}{p/2} = \frac{60 \times 50}{4/2} = 1500 \text{ rpm}$$

$$\text{slip} = \frac{N_s - N_r}{N_{sy}} \times 100\% = \frac{1800 - 1462}{1500} \times 100\% = 22.53\%$$



For a single phase:

$$Z_T = R_s + \frac{R_r}{s} + jX_{eq} = 15.60 \angle 7.4^\circ \Omega$$

$$I_s = \frac{V_{ph}}{Z_T} = 14.74 \angle -7.4^\circ \text{ A}$$

$$P_{gap} = I_s^2 \cdot \frac{R_r}{s} = 3.26 \text{ kW}$$

$$P_{mech} = P_{gap}(1-s) = 3.18 \text{ kW}$$

For 3 phase $P_{mech_total} = 3P_{mech} = 9.54 \text{ kW}$

$$(iii) T = \frac{P_{mech_total}}{\omega} = \frac{P_{mech_total}}{N_r \cdot \frac{2\pi}{60}} = 62.31 \text{ Nm}$$

$\approx 62.3 \text{ Nm}$

$$\approx 62.3 \text{ Nm}$$

(iV) For a single phase:

出力回線: $P_c = \frac{V_m^2}{R_c} = 88.17 \text{ W}$

$P_s = I_s^2 R_s = 47.77 \text{ W}$

$P_r = I_s^2 R_r = 82.56 \text{ W}$

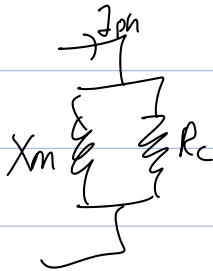
Losses
 $\Rightarrow P_{\text{loss}} = P_c + P_s + P_r$

効率 $\Rightarrow \eta = \frac{P_{\text{mech}}}{P_{\text{mech}} + P_{\text{loss}}} = 42.21\%$

Efficiency

3 phase or 1 phase?

(v) When no load:



for single phase:

$Z_T = jX_m // R_c = 145.52 \angle -76.0^\circ$

$\Rightarrow I_{ph} = \frac{V_{ph}}{Z_T} = 1.58 \angle -76.0^\circ$

$|I_{ph}| = 1.58 \text{ A}$